

Two new *PPX2* mutations associated with resistance to PPO-inhibiting herbicides in *Amaranthus palmeri*

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Abstract

BACKGROUND: Resistance to herbicides that inhibit protoporphyrinogen oxidase (PPO) is a widespread and growing problem for weed managers across the midwestern and midsouthern United States. In *Amaranthus* spp., this resistance is known to be conferred by a glycine deletion at the 210th amino acid (Δ G210) in PPO2. Preliminary analysis indicated that the Δ G210 mutation did not fully account for observed resistance to PPO inhibitors in two *Amaranthus palmeri* populations from Tennessee and one from Arkansas.

RESULTS: Sequencing *PPX2* cDNA from six resistant plants uncovered two new mutations at the R98 site (R98G and R98M), a site previously found to endow PPO-inhibitor resistance in *Ambrosia artemisiifolia*. Sequencing of this region from additional plants sprayed with 264 g fomesafen ha⁻¹ showed the presence of one or both R98 mutations in a subset of the resistant plants from all three populations. No plants sensitive to fomesafen contained either mutation. A derived cleaved amplified polymorphic sequence (dCAPS) assay to test for the presence of these mutations in *A. palmeri* was developed.

CONCLUSION: Two new mutations of *PPX2* (R98G, R98M) likely confer resistance to PPO-inhibitors in *A. palmeri*, and can be rapidly identified using a dCAPS assay.

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1 INTRODUCTION

Herbicides that inhibit protoporphyrinogen oxidase (PPO) have been used for about 50 years, primarily to control broadleaf weed species.¹ Over the past few years, there has been an increase in the use of PPO-inhibitors, driven primarily by widespread occurrence of glyphosate-resistant weeds, including *Amaranthus tuberculatus* (waterhemp) and *Amaranthus palmeri* (Palmer amaranth).^{2,3}

To date, 10 weed species have evolved resistance to PPO-inhibitors.⁴ The first of these, *A. tuberculatus*, was initially identified in Kansas in 2001.⁵ Since then, resistance to PPO-inhibitors in *A. tuberculatus* has been documented widely throughout the midwestern USA.⁴ The mechanism of resistance to PPO-inhibitors in *A. tuberculatus* was determined to be an insensitive site of action. Specifically, the loss of three consecutive nucleotides results in the deletion of a glycine residue at position 210 (Δ G210) of the PPO2 enzyme.⁶ The same mutation was subsequently found to be responsible for resistance in numerous other *A. tuberculatus* populations as well as in *A. palmeri*³ and, to date, no other PPO-inhibitor-resistance mechanism has been reported in either of these species.^{7–9}

Since the first report of an Arkansas *A. palmeri* population with resistance to PPO-inhibiting herbicides, additional populations have been reported in at least two other states (Tennessee and Illinois).⁴ Herein we describe our investigation of a new PPO-inhibitor-resistant population from Arkansas and two from

Tennessee. Initial analysis revealed that the Δ G210 mutation did not fully account for resistance to PPO-inhibitors in these populations. Through further analysis, we identified two additional mutations that most likely confer target-site resistance to PPO inhibitors.

2 EXPERIMENTAL METHODS

2.1 Plant material and growth conditions

Three *A. palmeri* populations with resistance to PPO inhibitors (R_{PPO}) were identified in Tennessee (Shelby County, labeled as 'SPW' and Lauderdale County, labeled as 'LPW') and Arkansas (Woodruff County, labeled as 'GPW') and compared to a PPO-inhibitor-sensitive (S_{PPO}) population from Tennessee (Knox

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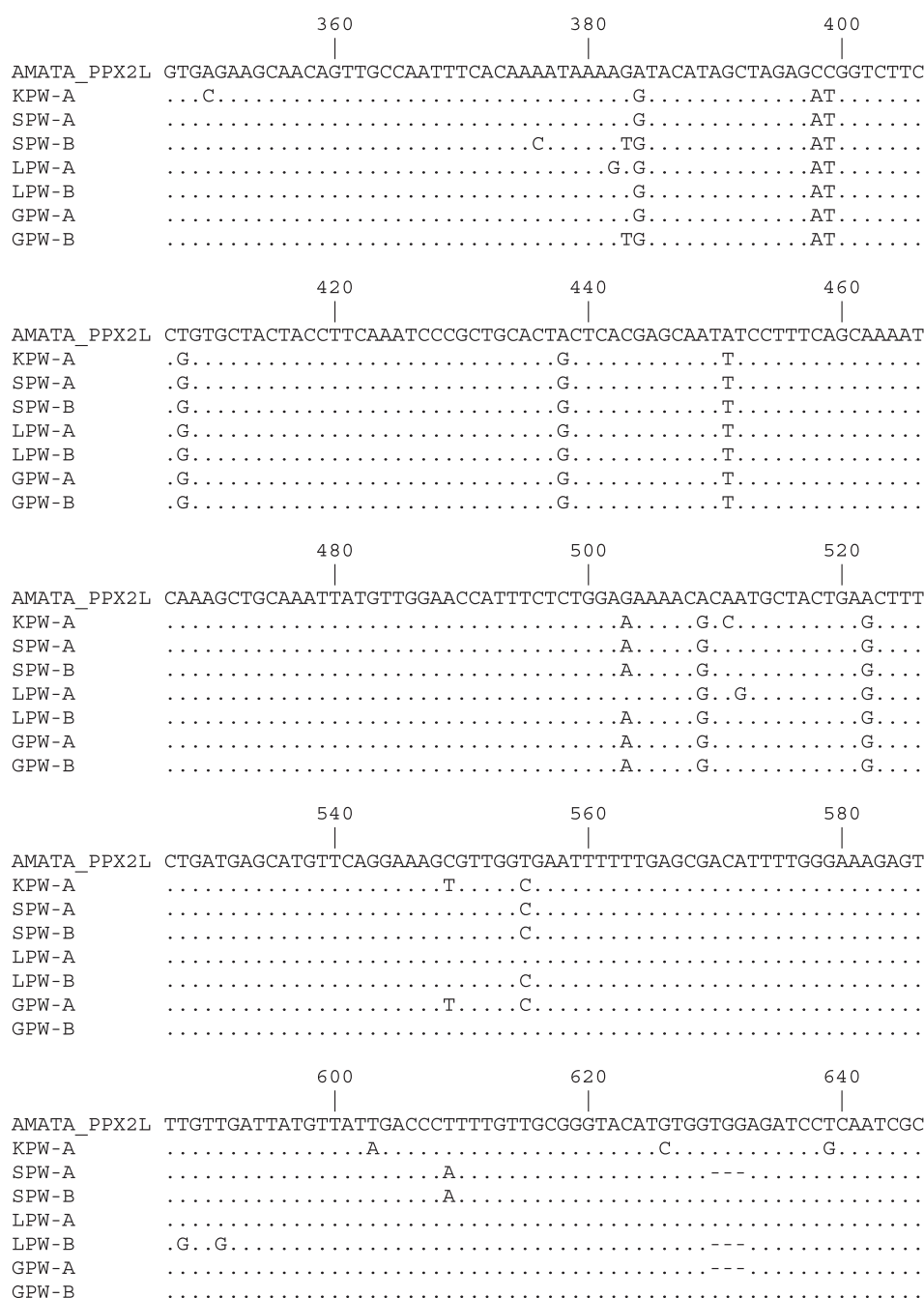


Figure 1. Alignment of partial cDNA sequences from two *Amaranthus palmeri* plants resistant to protoporphyrinogen-oxidase-inhibiting herbicides from each of three populations (SPW, LPW, and GPW) and one herbicide-sensitive plant (from population KPW) against the GenBank wild type *Amaranthus tuberculatus* sequence, DQ386117.1 (AMATA_PPX2L). Residues matching the consensus are marked with a dot (.) and deletions are marked with a dash (-). At the R98 site (382–384), both AGG (found in *A. palmeri*) and AGA (found in *A. tuberculatus*) encode an arginine (R) amino acid, whereas ATG and GGG encode a methionine (M) and glycine (G) amino acid, respectively. The G210 codon deletion occurs at nucleotides 630–632.

County, labeled as 'KPW'). Seeds were grown under greenhouse conditions as described previously.¹⁰

2.2 Herbicide treatment and sample collection

To select resistant plants for mechanistic studies, 18 plants from each population were sprayed with fomesafen (Reflex; Syngenta Crop Protection Inc., Greensboro, NC, USA) at a typical field use rate (264 g ai ha⁻¹) using a spray chamber¹⁰ when plants were 10–15 cm in height. Tissue was collected from each plant prior to spraying

and genomic DNA was extracted from both R_{PPO} and S_{PPO} plants using a modified CTAB protocol.¹¹

2.3 PPX2 ΔG210 assay

A modified version of a previously described TaqMan qPCR assay⁹ was used to test for the presence of the ΔG210 mutation. This assay uses allele-specific probes to determine whether a plant is wild type or heterozygous/homozygous for the ΔG210 mutation. The only difference between this new assay and the assay given in Wuerfell *et al.*⁹ is new primers that recognize both *A. palmeri* and

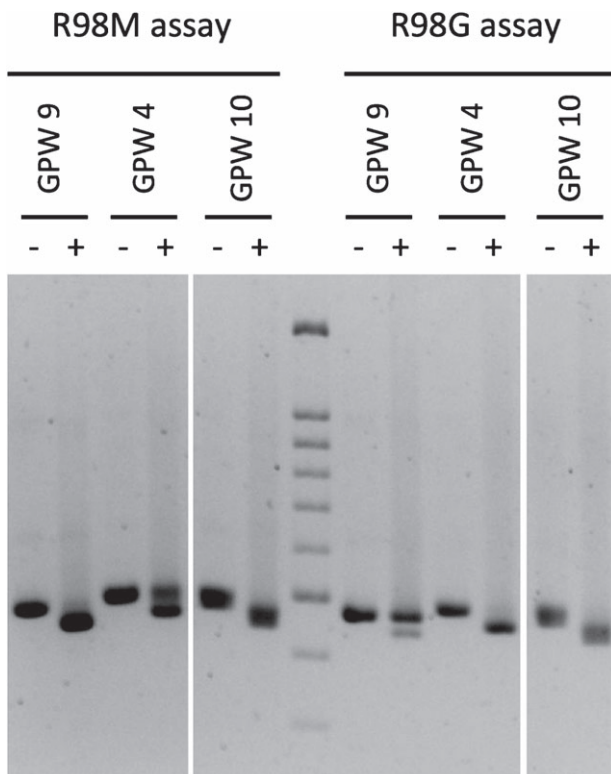


Figure 2. Gel image of dCAPS assays for three *Amaranthus palmeri* plants from population GPW. GPW 9 was identified as heterozygous for the R98G mutation, GPW 4 as heterozygous for the R98M mutation, and GPW 10 as wild type at R98 based on sequencing (Fig. 1). Both undigested (-) and digested (+) PCR products are shown. Additional plants were assayed, but a composite image is presented to show one representative sample from each genotype (wild type R98G, and R98M). The center lane is a DNA ladder; the PCR products fall between the 400-bp and 500-bp fragments.

A. tuberculatus PPX2 sequence, PA-tqF1 (5'-TGATTATGTTATTGACCTTTTGTGCG) and PA-tqR1 (5'-GAGGGAGTATAATTATTACAACCTCCAGAA).

2.4 Sequencing

To look for PPX2 mutations besides Δ G210, total RNA was extracted from six R_{ppo} plants and one S_{ppo} plant (TRIzol-based RNA extraction, Thermo Fisher #15596-026) and converted to cDNA (Proscript First Strand cDNA Synthesis Kit, NEB #E6300S). A nearly full-length PPX2 transcript was obtained by amplifying cDNA with the primers AmPPX2LpcF1 (5'-TCCATTACCCACCTTTCACC) and AmPPX2LspR (5'-TTACGCGGTCTTCTCATCCAT) and sequencing from both directions, using methods described previously.¹²

A second round of sequencing was performed on an arbitrarily selected set of the fomesafen-screened plants from GPW, SPW, LPW, and KPW. The R98 region (exons 4–6) of PPX2 was sequenced in at least 10 R_{ppo} and 10 S_{ppo} plants using the primers R98-F (5'-CTTGGGATACGTGAGAAGCAACAGTTG) and AmPPX1southernR (5'-AGCGGGATTGAAGGTAGTAG).¹²

2.5 dCAPs assay

Rapid genotyping of *A. palmeri* plants for presence or absence of the PPX2 R98 mutations was accomplished by developing a derived cleaved amplified polymorphic sequences (dCAPS) assay. Since two mutations at two different nucleotide positions within

Table 1. PPX2 mutations and herbicide responses among individual plants of three resistant populations (GPW, LPW, and SPW) and one sensitive population (KPW)

Plant	Δ G210 deletion ^a	R98 mutation ^b	Herbicide response ^c
GPW 1	Het	R98G	R
GPW 2	—	—	R
GPW 3	—	—	R
GPW 4	Het	R98M	R
GPW 5	—	R98G	R
GPW 7	Het	R98M	R
GPW 8	Het	R98M	R
GPW 9	Het	R98G	R
GPW 10	Het	—	R
GPW 11	Homo	—	R
GPW 13	—	R98G, R98M	R
GPW 14	—	R98G*	R
GPW 17	Het	R98G	R
GPW-A ^d	Het	—	R
GPW-B	—	R98M	R
LPW 1	—	R98G	R
LPW 2	Homo	—	R
LPW 3	Het	—	R
LPW 7	—	R98G	R
LPW 8	—	R98G	R
LPW 9	Het	—	R
LPW 12	Het	—	R
LPW-A	—	R98G	R
LPW-B	Het	—	R
SPW 1	—	—	R
SPW 7	—	—	R
SPW 13	—	R98G	R
SPW-A	Het	—	R
SPW-B	—	R98M	R
KPW 1	—	—	S
KPW-A	—	—	S
LPW 4	—	—	S
LPW 16	—	—	S
LPW 17	—	—	S
SPW 2	—	—	S
SPW 3	—	—	S
SPW 4	—	—	S
SPW 5	—	—	S
SPW 6	—	—	S
SPW 9	—	—	S

^a Plants were scored as homozygous (Homo) or heterozygous (Het) for the G210 deletion mutation, or as not having this mutation (—).

^b Plants were scored for the presence of a mutation conferring an R98G or R98M substitution, or as not having a mutation at this codon (—). The R98 mutations were all heterozygous, except where indicated (*).

^c Plants were scored as resistant (R) or sensitive (S) based on whether or not they survived fomesafen applied at 264 g ha⁻¹.

^d Plants labeled with a letter following the population name correspond to those used for cDNA sequencing (Fig. 1).

the PPX2 sequence were found, two dCAPS assays were created. Both assays required a nested PCR approach, with an initial amplification of a gene fragment using the AmPPX2LpcF1 and AmPPX2LspR1 primers described above, followed by a second amplification using the dCAPS primers. The R98M mutation was detected using the dCAPS primers R98-F (5'-CTTGGGATACGTGAAGCAACAGTTG) and R98M-R (5'-TAGCAACGGAAGACCATCTCTATCTAGGTAC) while the same forward primer (R98-F) was used in

conjunction with a second reverse primer, R98G-R (5'-TAGCAACG-GAAGACCATCTCTATGAGC), to detect the R98G mutation. The resulting PCR products were mixed with 1 unit of either restriction enzyme (KpnI-HF for R98M and HindIII-HF for R98G, NEB #R3142S and #R3104S) into 1× CutSmart Buffer and digested for 4 h at 37°C. Any fully digested products were scored as wild type and partially digested or non-digested products were scored as heterozygous or homozygous mutants, respectively.

3 RESULTS AND DISCUSSION

3.1 *A. palmeri* PPX2 sequencing

cDNA sequencing from one S_{PPO} and six R_{PPO} plants revealed the presence of two new mutations at the R98 residue, one conferring an R98G and the other an R98M amino acid substitution (Fig. 1). A mutation at the homologous site previously was found to confer resistance to PPO-inhibitors in *Ambrosia artemisiifolia* but, in this case, the mutation conferred an R98L substitution.¹³ All six of the R_{PPO} plants had either the $\Delta G210$ codon deletion common in *Amaranthus* species or one of the R98 mutations.

It was previously argued that the R98L mutation observed in *A. artemisiifolia* was unlikely to occur in *A. tuberculatus*, because the wild type codon (AGA) of the latter would require two mutations to be converted to a leucine codon.¹³ Similarly, the codon (AGG) observed in the sensitive *A. palmeri* plant would require two mutations to be converted to a leucine codon, providing an explanation for why R98G and R98M amino acid substitutions were selected instead (each requiring only one nucleotide change). The R98M amino acid substitution observed in *A. palmeri* would require two mutations in the known AGA wild type codon in *A. tuberculatus*, providing an explanation for why it has not been observed in *A. tuberculatus* (although it is possible that an AGG or other arginine codon may also occur frequently as the 'wild type' in *A. tuberculatus*). In contrast, only one mutation is needed to convert the *A. tuberculatus* AGA codon to a glycine codon, raising the question as to why the R98G mutation observed here in *A. palmeri* has not yet been reported from *A. tuberculatus*.

3.2 Co-segregation of R98G and R98M with fomesafen resistance

All 18 plants analyzed from the KPW sensitive population were killed by the applied rate of fomesafen. Plants from the three resistant populations were divided into two main categories: (1) plants that responded similarly to KPW plants, and (2) plants that exhibited initial damage followed by regrowth and survival (data not shown). Further sequencing and assaying of fomesafen-screened plants found the presence of the $\Delta G210$, R98M, and/or R98G mutations in R_{PPO} plants only (Table 1). In a few cases (four out of 29), an R_{PPO} plant showed none of the three mutations, perhaps indicating yet another mechanism of PPO-inhibitor resistance in these populations. Many plants had both the codon deletion and an R98 mutation, but it is not known if they occur on the same allele. Disregarding plants with the $\Delta G210$ mutation (which is known to be sufficient for resistance⁶), all ten plants that had either the R98G or R98M mutation were resistant, and none of the eleven sensitive plants analyzed had either of these mutations, indicating that either of the R98 substitutions is sufficient to confer fomesafen resistance. Although there were no apparent differences between the fomesafen responses of plants with the R98G, R98M, or $\Delta G210$ mutations, crosses are underway to create plants homozygous for the R98G and R98M mutations so that these resistant biotypes can be more fully characterized.

3.3 dCAPS assay

The dCAPS assays were successful in identifying heterozygous and homozygous mutants for both R98 substitutions (Fig. 2). These markers, together with the $\Delta G210$ TaqMan-based assay, can be used to aid in rapid screening of *A. palmeri* plants for resistance to PPO inhibitors. The newly developed $\Delta G210$ TaqMan assay worked for both *A. palmeri* and *A. tuberculatus* (data not shown). The R98G dCAPS assay should work for any species with either the AGA or AGG arginine codon at the R98 site because a mismatch for the restriction enzyme site was created in the reverse primer corresponding to the third nucleotide of the codon. However, this assay would improperly score an AGT or AGC serine codon as being wild type. For the R98M assay, the 3' nucleotide (C) of the primer anneals to the third nucleotide of the R98 codon. If the third nucleotide deviates from the wild type (G; observed in all 10 S_{PPO} plants sequenced) the primer may or may not amplify.

4 CONCLUSIONS

In three *A. palmeri* populations, two from Tennessee and one from Arkansas, two new mutations in PPX2 are associated with PPO-inhibitor resistance. The co-segregation of the R98 mutations with fomesafen resistance plus the highly-conserved nature of the R98 site for substrate binding¹⁴ are good evidence for the involvement of these mutations in PPO-inhibitor resistance. This is the first report of an R98 mutation in *Amaranthus* spp. and a surprising discovery given the prevalence of the $\Delta G210$ in these species. The fact that two R98 mutations evolved alongside the widespread $\Delta G210$ mutation speaks to the intense selection pressure these weed populations are under.

The dCAPS assay described above should be useful for rapid detection of PPO-inhibitor resistance in *A. palmeri*. The current assay to screen only for the $\Delta G210$ codon deletion misses a substantial number of resistant plants. By quickly confirming that PPO-inhibitor resistance is prevalent in a field, alternative weed management strategies can be implemented to effectively manage the resistant population. Further research is needed to more thoroughly characterize the new resistant biotypes identified herein, and to determine if additional mechanisms conferring resistance to PPO inhibitors are prevalent in *A. palmeri*.

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